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Started	Monday, 26 October 2026, 1:28 PM
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Duration	7 mins 48 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search requires the list to be sorted.
- c. Linear search is more efficient than binary search on large lists.
- d. Linear search divides the list into halves.

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 99 ✓
- b. 90 and 94
- c. 89 and 99
- d. 89 and 94

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By sorting the list first
- b. By using a hash function
- c. By comparing each element in the list sequentially ✓
- d. By dividing the list into halves

Question **4**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search starts over from the beginning
c. The search continues until the end of the list
d. The search stops immediately ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list
b. Divide the list into two equal parts
c. Compare the target element with the first element in the list
d. Compare the target element with the middle element in the list ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is large and unsorted
c. When the list is small and sorted
d. When the list is small and unsorted ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array
b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
c. There must be a mechanism to access middle element directly
d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Hash search
c. Linear search ✓
d. Both (1) & (3)

Question **9**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search can find multiple instances of the target element
c. Binary search does not require dividing the list
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **10**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(1) ✓
b. O👎
c. O(log n)
d. O(n log n)

Question **11**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. At the beginning of the list
c. In the left sublist
d. In the middle of the list

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The list is sorted

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is sorted ✓
c. When the list is unsorted
d. When the list is very small

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. An error is raised
c. The search is considered unsuccessful ✓
d. The last element is returned

Question **15**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓
b. log n
c. 1
d. n/2